

# Every Algebraic Number is a Critical Value of an Integer Polynomial

by RJ Acuña

## Abstract

For every irreducible  $f \in \mathbb{Z}[x]$  of positive degree we construct  $g \in \mathbb{Z}[x]$  and an irreducible  $F \in \mathbb{Z}[x]$  with  $F \mid g'$  and  $F^2 \mid f \circ g$ . Equivalently, every algebraic number is a critical value of an integer polynomial, so the set of critical values of  $\mathbb{Z}[x]$  is exactly  $\overline{\mathbb{Q}}$ . The construction is explicit. The proof was assisted by AI, and has been formalized in Lean.

## 1 Introduction

A critical value of a nonconstant  $g \in \mathbb{Z}[x]$  is a number  $\alpha$  with  $g(\beta) = \alpha$  and  $g'(\beta) = 0$  for some  $\beta$ . Since  $\overline{\mathbb{Q}}$  is closed under polynomial evaluation, every critical value is algebraic. Johannes Walcher posed the problem of establishing the reverse implication. This note solves Walcher's problem with an explicit construction.

**Theorem 1.** *Let  $f$  be an irreducible integral polynomial of positive degree. Then there exist  $F$  and  $g$  in  $\mathbb{Z}[x]$ , with  $F$  irreducible and of positive degree, such that*

$$F \mid g' \quad \text{and} \quad F^2 \mid f \circ g.$$

The corollary is the statement in the title.

**Corollary 1.** *The set of critical values is exactly  $\overline{\mathbb{Q}}$ .*

**pf.** Write  $f = m_\alpha$ . If  $g(\beta) = \alpha$  and  $g'(\beta) = 0$  then  $\beta$  is a double root of  $f \circ g$ , since  $(f \circ g)' = f'(g) g'$ ; so  $m_\beta \mid g'$  and  $m_\beta^2 \mid f \circ g$ . Conversely take  $F$  and  $g$  from Theorem 1: a root  $\beta$  of  $F$  has  $g'(\beta) = 0$  and  $f(g(\beta)) = 0$ , so  $g(\beta)$  is a conjugate of  $\alpha$ , and conjugation gives  $\alpha$  itself. ■

## 2 The construction

Let  $f$  be irreducible of degree  $n \geq 1$  and  $\alpha$  a root. Write  $f(x) = \sum_{k=0}^n a_k x^{n-k}$ , so that  $a_n = f(0)$ . If  $f = x$ , then  $g = x^2$  and  $F = x$  work. Otherwise  $f(0) \neq 0$ , and since  $\gcd(f, f') = 1$  in  $\mathbb{Q}[x]$ , the elimination ideal  $(f, f') \cap \mathbb{Z}$  is nonzero; let  $u, v \in \mathbb{Z}[x]$  and  $\rho \in \mathbb{Z}, \rho \neq 0$ , be such that  $u f + v f' = \rho$ , and set  $M := \rho a_n$ ,  $F(x) := f(Mx)/a_n$  and  $W(x) := (v(0)F(x) - v(Mx))/\rho$ . The polynomial claimed by Theorem 1 is

$$g(x) := Mx + f(Mx)W(x). \tag{1}$$

**Theorem** (restatement of Theorem 1).  *$F$  is irreducible, and  $F \mid g'$  and  $F^2 \mid f \circ g$ .*

**pf.** Write  $F(x) = \sum_{k=0}^n A_k x^{n-k}$ . From  $A_k = a_k M^{n-k}/a_n$  we get  $A_n = 1$  and  $A_k = a_k \rho^{n-k} a_n^{n-k-1}$  for  $k < n$ , so  $F \in \mathbb{Z}[x]$  with  $F(0) = 1$ , which gives  $W(0) = 0$ . And  $\rho$  divides  $M$  and every  $A_k$  with  $k < n$ , so it divides  $v(0)F(x) - v(Mx)$  and  $W \in \mathbb{Z}[x]$ . Therefore  $g$  is an integral polynomial.

Let  $\beta = \alpha/M$ , a root of  $F$ ; therefore  $g(\beta) = \alpha$ , and  $\rho W(\beta) = -v(\alpha)$ . Then

$$\begin{aligned} g'(\alpha/M) &= M + \rho a_n f'(\alpha) W(\alpha/M) + \cancel{f(\alpha)} W'(\alpha/M) \\ &= M - a_n f'(\alpha) v(\alpha) = 0, \end{aligned}$$

because  $u(\alpha) \cancel{f(\alpha)} + v(\alpha) f'(\alpha) = \rho$ . Hence  $\beta$  is a double root of  $f \circ g$ , and  $F$  is primitive irreducible, so  $F \mid g'$  and  $F^2 \mid f \circ g$  by Gauss. ■

Any  $\rho$  will do, but  $M = \rho a_n$  multiplies into every coefficient of  $F$ ,  $W$  and  $g$ , so the reduced resultant gives the answer with the smallest coefficients.

### 3 Examples

Each line below is a closed identity, checkable by expansion.

*Example 1* ( $f = qx - p$ ,  $p \neq 0$ ,  $\gcd(p, q) = 1$ ). Here  $f' = q$  and  $p = qx - f$ , so  $(f, f') \cap \mathbb{Z} = \mathbb{Z}$  and  $\rho = 1$ . Choose  $a, b \in \mathbb{Z}$  with  $ap + bq = 1$  and take  $u = -a$ ,  $v = ax + b$ ; then  $f(0) = -p$ ,  $M = -p$ ,  $F = qx + 1$ ,  $W = x$ , and

$$g = -pqx^2 - 2px, \quad g' = -2p(qx + 1), \quad f \circ g = -p(qx + 1)^2.$$

*Example 2* ( $f = ax^2 + bx + c$ ). This is irreducible of degree 2 exactly when  $a \neq 0$ ,  $\gcd(a, b, c) = 1$ , and  $D := b^2 - 4ac$  is not a square. Since  $(f')^2 = 4af + D$ ,

$$\rho = \text{Res}(f, f') = -aD, \quad u = 4a^2, \quad v = -af',$$

so  $M = -acD$  and

$$F = a^3cD^2x^2 - abDx + 1, \quad W = a^3bcDx^2 - a(b^2 - 2ac)x,$$

$$g = a^6bc^3D^3x^4 - 2a^4c^2(b^2 - ac)D^2x^3 + a^2bc(b^2 - ac)Dx^2 - 2ac(b^2 - 3ac)x,$$

with  $g' = 2ac(2a^2bcDx - (b^2 - 3ac))F$ . For  $b = 0$  the quartic term drops and  $\deg g = 3 = n + 1$ , the least any solution can have.

*Example 3* ( $f = x^3 - x - 1$ ). The construction gives  $M = 23$  and a sextic. A hand-tuned quintic also witnesses the *statement*, and is a useful independent check of it:

$$g = -24334x^5 + 39675x^4 - 26450x^3 + 9085x^2 - 1610x + 118, \quad F = 23x^3 - 23x^2 + 8x - 1,$$

with  $g' = -230(23x - 7)F$  and  $F^2 \mid f \circ g$ , the cofactor having degree 9.

*Example 4* (Dedekind's cubic).  $f = x^3 - x^2 - 2x - 8$ , the standard example of a field whose ring of integers is not monogenic: 2 is a common index divisor [Ded78]. With  $u = 21x - 125$ ,  $v = -7x^2 + 44x - 3$ ,  $\rho = 1006$ ,  $f(0) = -8$ ,  $M = -8048$ ,

$$F = 65158925824x^3 + 8096288x^2 - 2012x + 1, \quad W = -194310912x^3 + 426544x^2 + 358x,$$

$$g = 101288722414414331904x^6 - 209759614012620800x^5 \\ - 217370176548864x^4 - 14767629312x^3 + 2350016x^2 - 10912x.$$

### 4 Formalization

The development above has been formalized in Lean 4 [MU21] against Mathlib [The20], in about 1,200 lines. The main statement is

```
1 theorem main (f : Z[X]) (hf : 1 <= f.natDegree) :
2   exists g H : Z[X], Irreducible H /\ H \mid derivative g /\ H^2 \mid f.comp g
```

with a strengthened form carrying  $2 \leq \deg g$  and  $1 \leq \deg H$ , and with Corollary 1 proved from it. Every result depends only on the three standard axioms `propext`, `Classical.choice` and `Quot.sound`.

## References

- [Ded78] Richard Dedekind. “Über den Zusammenhang zwischen der Theorie der Ideale und der Theorie der höheren Kongruenzen”. In: *Abhandlungen der Königlichen Gesellschaft der Wissenschaften zu Göttingen* (1878).
- [The20] The mathlib Community. “The Lean Mathematical Library”. In: *Certified Programs and Proofs (CPP 2020)*. ACM, 2020.
- [MU21] Leonardo de Moura and Sebastian Ullrich. “The Lean 4 Theorem Prover and Programming Language”. In: *Automated Deduction – CADE 28*. Lecture Notes in Computer Science. Springer, 2021.